

Pruritus

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Definition:

- Itching of the skin, an unpleasant sensation that triggers rubbing or scratching.
- The sensation of pruritus can be extremely disturbing, leading to disrupted sleep and even depression.
- Localized or generalized.
- May be with or without skin changes.
- Acute or chronic.

Skin comes in contact with allergens or irritating substances



Irritation and inflammation of epidermal nerve endings
(unmyelinated C, type A delta nociceptive)



Release of histamine



More itching



More scratching

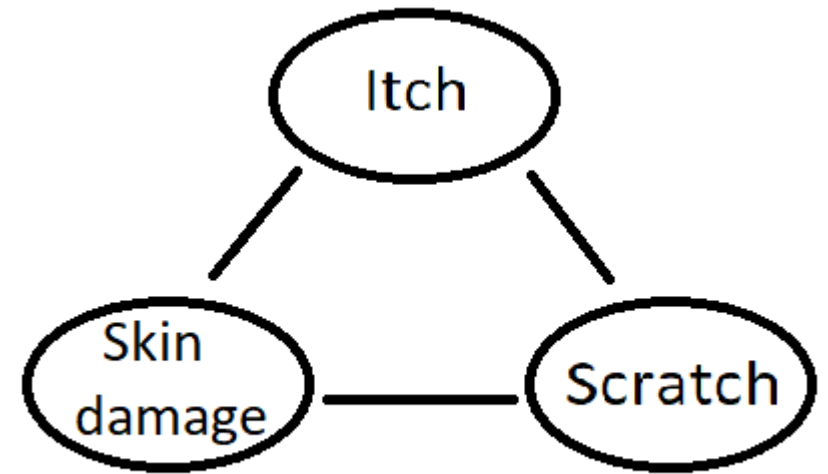


More damage and inflammation



Itch scratch cycle continues

Pathogenesis:



Complications:

- Skin injury.
- Infections; viral, bacterial, fungal.
- Scarring and pigmentary changes.



Classification:

- 1. Pruritus with skin changes.**
- 2. Pruritus with normal skin.**

Pruritus with skin changes

Could be localized or generalized

Localized pruritus with skin changes:

- Eczema
- Psoriasis
- Lichen planus (Polygonal purple pruritic flat-topped (planar) itchy papules and plaques of unknown cause)
- Dermatitis herpetiformis (gluten allergy with rash, characteristically excoriated papulovesicular eruption on elbows and buttocks, associated with GI symptoms like bloating sometimes)



Dermatitis herpetiformis



Lichen Planus



Psoriasis



Atopic dermatitis

Localized pruritus with skin changes:

- Insect bites/stings, (nodular prurigo may develop after insect bites and is characterized by persistent itching, lichenified papules, and nodules)
- Head lice
- Contact dermatitis
- Polymorphic light eruption (an acute allergy to sunlight on sun-exposed skin)



Insect bites



Polymorphic light eruption



Contact dermatitis



Insect bites



Prurigo Nodulris

Localized pruritus with skin changes:

- Urticaria or angioedema ('hives' with swelling, especially of the face)
- Fungal infections (particularly tinea pedis of the feet)
- Pruritus ani (perianal itching, a common condition that may result from anal leakage, skin tags, hemorrhoids, thread worms, excessive washing, the use of medicated wipes or allergy to hemorrhoid creams containing Balsam of Peru)
- Pruritus vulvae (intense itching may result from lichen sclerosus et atrophicus, Candida infections, or eczema).



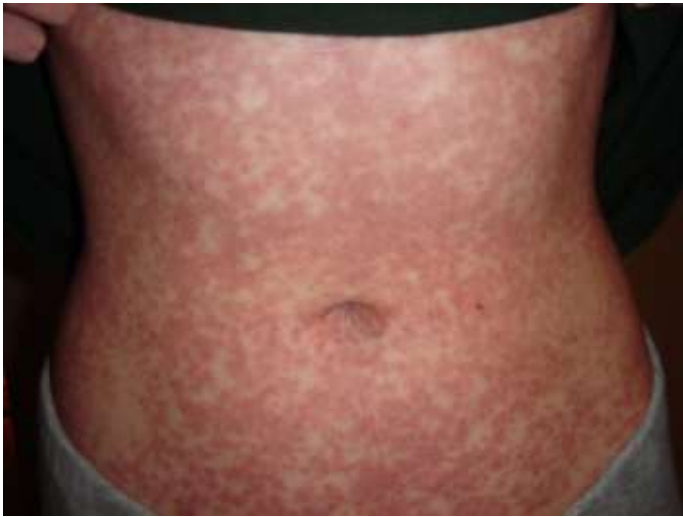
Tinea Pedis



Urticaria

Generalized pruritus with skin changes:

- Widespread eczema/psoriasis.
- Scabies.
- Allergic drug eruptions (antibiotics and anticonvulsants)
- Graft versus host disease (following bone marrow transplantation).



Allergic drug eruptions



Graft versus host disease



Scabies

Generalized pruritus with skin changes:

- Pre-bullous pemphigoid (dermatitic eruption before blisters appear).
- Cutaneous lymphoma (may start over the buttocks as more localized disease).
- Parasitophobia (a belief that there are parasites under the skin, excoriations are seen).
- Body lice or pubic lice (body lice live in the clothing).



Pre-bullous pemphigoid



Cutaneous lymphoma



Body lice

Generalized pruritus with skin changes:

- Viral exanthems (rashes associated with systemic viral illness)
- Xerosis (dry skin, especially in the elderly).



Xerosis



Viral exanthems

Pruritus with skin changes:

- Investigation and management of pruritus with skin changes should be directed at the suspected underlying cause.

Pruritus with normal skin

Pruritus with normal skin:

- Medical practitioners need to be alert to the patient with generalized pruritus and normal skin as itching may be the first symptom of a systemic disorder

Systemic causes:

- **Endocrine** – diabetes, myxoedema, hyperthyroidism, menopause, and pregnancy.
- **Metabolic** – hepatic failure, biliary obstruction and chronic renal failure.
- **Haematological** – polycythemia and iron deficiency anemia.
- **Malignancy** – lymphoma, leukemia, myeloma, and carcinomatosis.
- **Neurological/psychological** – neuropathic pruritus, multiple sclerosis, and anxiety.
- **Infection** – filariasis, hookworm, and HIV.
- **Drugs** – opioids.

Investigations of pruritus with normal skin:

- **Blood tests:**

1. Full blood count, erythrocyte sedimentation rate, liver, and renal function.
2. Serum iron, ferritin, and total-iron binding capacity.
3. Thyroid function and fasting glucose.
4. Serum protein electrophoresis.
5. HIV antibodies.

- **Urine analysis.**

- **Stools for blood and parasites/ova.**

- **Chest X-ray.**

- **Skin biopsy for direct immunofluorescence.**

Table 1. Historical Findings That Suggest Etiologies for Pruritus

<i>Historical finding</i>	<i>Possible etiologies</i>
New cosmetics or creams	Allergic contact dermatitis, urticaria, photodermatitis
New medications, supplements, or illicit drugs	Urticaria, fixed drug eruptions
Recent travel	Pediculosis, scabies infestation, photodermatitis, urticaria
Hobby or occupational exposure to solvents, adhesives, cleaners	Irritant contact dermatitis, xerosis, atopic dermatitis, eczema
New animal exposures	Flea infestation, allergic contact dermatitis, urticaria
Sick contacts, especially those with febrile diseases and rashes	Rubeola, mumps, varicella, scarlet fever, cellulitis, fifth disease, folliculitis
Unexplained weight changes, menstrual irregularity, heat/cold intolerance	Thyroid disease with secondary urticaria or xerosis
Unexplained weight loss, night sweats, unexplained fevers, fatigue	Lymphoma with secondary generalized pruritus
Malaise, nausea, decreased urine output	Renal failure with generalized pruritus

Management:

- Identifying and treating the underlying cause.
- Patients themselves will find some short-term relief by scratching the skin, but this ultimately leads to further itching and scratching, the so-called ‘itch-scratch cycle’.
- To break this cycle the sensation of itch needs to be suppressed, or the patient’s behavior changed by **habit reversal techniques**.
- Individuals can purchase simple soothing emollients over the counter and apply these frequently to the pruritic areas.
- The soothing effect can be enhanced by storing the cream in the fridge and applying them cold.

Management:

Topical treatment:

- **Menthol**
- **Camphor**
- **Crotamiton**
- **Doxepin**
- **Topical local anesthetics (containing benzocaine, lidocaine, or tetracaine)**
- **Topical antihistamines (mepyramine and antazoline)**

Management:

Systemic treatment:

- Oral antihistamines: non-sedating during the day and sedating ones at night. Usually, we use histamine H1 blockers cetirizine, levocetirizine, desloratadine, and fexofenadine during the day and hydroxyzine at night.
- H2 blockers (ranitidine and cimetidine) may be used in addition to H1 blockers in resistant cases.

Management:

Pruritus ani/vulvae:

- The affected area should be washed once daily
- Excessive washing should be avoided
- Aqueous cream or emulsifying ointments can be used as a soap substitute.
- Patients should avoid perfumed/colored/medicated toilet tissue or wipes.
- Simple paraffin or zinc cream can be used as a barrier ointment to prevent skin irritation from anal leakage/vaginal discharge.
- **Weak topical steroids** can help reduce inflammation and itching.
- If symptoms persist, patients should see a dermatology specialist who may do **patch testing** (contact dermatitis) or perform **a skin biopsy** (neoplasia, lichen sclerosus or lichen planus).

Thank you